Dominica: Public Debt Sustainability Analysis

1. This section assesses the sustainability of Dominica's public debt using the debt sustainability analysis (DSA) template for low-income countries. The DSA was conducted jointly by Fund and Bank staff.

2. Based on the analysis in the baseline scenario and several alternative scenarios, the staff conclude that while Dominica's public debt is sustainable at current output growth rates and fiscal stance, there are risks of future debt distress if these variables weakened. In addition, the deficits that Dominica's social security system (DSS) is expected to face starting in 2014 pose another challenge to the country's public finances.

Dominica's public debt situation at end-2004

3. Dominica initiated a collaborative debt restructuring process in 2004 with a view to reducing its debt to a more sustainable level. Following discussions with creditors, the restructuring targeted an NPV to face value reduction of 50 percent. So far, creditors with about 70 percent of the debt eligible for restructuring have agreed to the proposal, and the authorities remain in good faith negotiations with the remaining hold-out creditors.

4. It is assumed that those creditors not currently participating in the restructuring receive the intermediate bond. In addition, external interest arrears are excluded since arrears with participating creditors have been settled as part of the debt restructuring and arrears with nonparticipating creditors are either in dispute or expected to be settled when an agreement is reached. Undisputed interest arrears amount to approximately 0.4 percent of GDP.

5. As of end of 2004 and after taking into account the restructuring, Dominica had a public debt of around 116 percent of GDP, with external public debt reaching around 81 percent of GDP. The net present value to GDP ratios stood at approximately 90 percent for the total public debt and 54 percent for the external. Around 47 percent of GDP is owed to multilateral creditors (with the Caribbean Development Bank holding almost two thirds of this). External debt with bilateral creditors is approximately 19 percent of GDP while debt to commercial creditors is around 16 percent of GDP). With respect to domestic debt, about 15 percent of GDP is owed to Dominica's Social Security System (DSS) and 20 percent of GDP is owed to commercial banks, the ECCB, and corporations.

The baseline scenario

6. The baseline scenario is developed for 2005–2025 and refers to the public debt of the nonfinancial public sector. The main assumptions are:

- GDP growth stays at 3 percent. While this assumption implies a rate of growth higher than the average observed in the 1990s (2 percent), it seems consistent with the stronger growth observed in 2004 and first half of 2005 and with growth recorded before the 1990s. In addition, this rate of growth seems achievable given the

improved macro policy environment, as well as the structural reforms envisioned in the draft Growth and Social Protection Strategy document (which is close to being finalized) and the favorable prospects for the tourism sector.

- The primary balance of the central government remains at 3 percent of GDP over the projection period, while pubic enterprises run an overall deficit of 0.5 percent of GDP. The assumption about the government primary balance is consistent with the strong fiscal turnaround Dominica has had in recent years and with the target that the government has committed to maintain under the program. The assumption on public enterprises follows the average observed during the period 1999–2004.

- Annual disbursements of external concessional debt reach 1.5 percent of GDP, which is consistent with the projections of the country's public sector investment program (PSIP).

- New domestic financing can be obtained by the central government and public enterprises at an interest rate of 7 percent.

7. Under the baseline scenario (Table 1a), all the indicators of debt burden show that Dominica's debt remains sustainable. The only indicator that does not decline continuously is the debt service to revenue ratio, which temporarily shows an upward trend and then a decline starting in 2014.[20] While the increase in the ratio of debt service to revenue is temporary, the high debt servicing costs point to the need of maintaining fiscal discipline in order to avoid liquidity constraints.

8. Under this scenario, Dominica would reach a public debt to GDP ratio of 60 percent by 2018.

Sensitivity analysis

Changes in growth and primary balance scenarios

9. The sensitivity analysis illustrates two important points (Table 2a). First, if Dominica primary balance and economic growth return to their averages of the last ten years (a 1 percent of GDP primary deficit and 0.8 percent growth), then public debt turns unsustainable (case described in Scenario A.1). Second, if Dominica can keep the fiscal effort it made during last year (a primary surplus of around 4.3 percent of GDP), then public debt remains sustainable (Scenario A.2).

[20] The initial upward trend is mainly caused by two features of the debt restructuring. First, the grant extended by the CDB covers interest payments and amortization of several of the loans Dominica has with that institution in 2005 and 2006, which makes the debt service to GDP low in those years. Second, Dominica's debt restructuring includes a repurchase clause which is expected to increase amortization starting in 2009.

10. In addition, the sensitivity analysis shows the importance of sustaining growth at current levels. In the alternative Scenario A.3 in which growth falls to 2½ percent per year, most indicators of debt burden present a slight upward trend. Behind this adverse effect of growth on the debt burden is the fact that as output growth slows, fiscal revenues would also likely decline, causing the primary balance to deteriorate. As the interest rate on new nonconcessional financing (7 percent) is much larger than Dominica's current average interest rate (around 2.8 for external debt and 5 percent for domestic debt), this scenario also increases interest expenditures over time.

Social Security

11. Dominica Social Security (DSS) is currently running surpluses, but deficits are expected to emerge starting in 2014 and grow continuously over time. At the request of the Dominican authorities, a World Bank/FAD technical assistance mission visited the country in June 2005 to assess the financial situation of DSS and propose possible reforms.

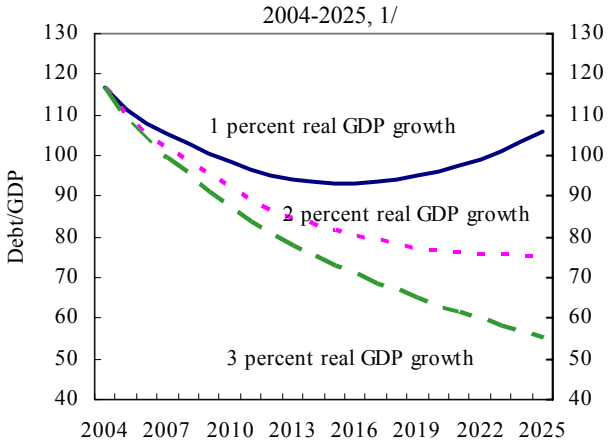
12. While the projected deficits in the social security system are expected to be addressed through structural reforms, a scenario on the implications of direct transfers from the central government on the debt path in absence of reforms was also considered. The exercise assumes that the government starts making transfers in 2014 in order to cover DSS' deficits. Under this scenario, public debt to GDP initially declines (reaching around 60 percent in 2020) but then starts to increase again as DSS deficits become larger over time.

Changes in interest rates

13. Given that Dominica's debt to GDP ratio is still high, the implication of adverse changes in interest rates was also considered.

14. Chart 1a describes the path of the public debt to GDP ratio assuming the interest rates available for new financing reaches 12 percent (5 percentage points above the baseline scenario) and different growth rates. As can be seen in the graph, public debt remains sustainable when the other key variables remain under the baseline assumptions (i.e., 3 percent GDP growth and 3 percent primary surplus) although its decline is much slower than in the baseline scenario. When GDP growth declines to 2 percent, then public debt stabilizes at around 75 percent of GDP at the end of the projection period. With a GDP growth of 1 percent, public debt would return to an unsustainable path.

Chart 1.a. Dominica: Public Sector Debt to GDP 2004-2025, 1/



1/ Assumes 3 percent primary surplus by the Central Government. Excludes the impact of DSS deficits and abstracts from natural disasters. The interest rate for new financing is assumed to be 12 percent, which is five percentage points above the baseline assumption.

Natural disasters

15. The impact of natural disasters can be analyzed using the stress tests provided in Table 2a. For instance, the fifth bound test presented in Table 2a suggests that Dominica's debt would remain sustainable after a natural disaster that increased the debt to GDP ratio by 10 percent in 2006.[21]

[21] Given the variability of the impact of natural disasters, it is difficult to determine the appropriate magnitude for this test. However, the ECCU Selected Issues Paper from 2004, looks at a sample of 12 large natural disasters occurring in the ECCU since 1970 and reports a median public debt increase of 6.5 percent of GDP over a period of three years after the disaster. The shock presented here is larger than what would be suggested by the previous evidence.

Table 1a.Dominica: Public Sector Debt Sustainability Framework, 2002-2025
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2002	2003	2004	Historical Average 5/	Standard Deviation 5/	2005	2006	2007	2008	2009	2010	Average 2005-2010	2015	2025	Average 2011-2025
Public sector debt 1/	127.2	130.6	116.7			109.1	103.9	99.5	95.3	91.1	87.1		69.4	39.7	
o/w foreign-currency denominated 7/	79.9	84.4	81.2			81.1	77.6	72.9	68.5	64.2	58.7		39.7	23.1	
Change in public sector debt	33.6	3.4	-13.9			-7.6	-5.2	-4.4	-4.3	-4.2	-4.0		-3.3	-2.9	
Identified debt-creating flows 8/	11.0	-2.6	-11.6			-7.6	-5.0	-4.4	-4.3	-4.2	-4.1		-3.3	-3.0	
Primary deficit	1.2	-6.1	-4.4	0.9	4.5	-3.3	-3.3	-3.3	-3.3	-3.2	-3.2	-3.3	-3.1	-3.1	-3.1
Revenue and grants	41.2	49.9	48.8			47.5	47.3	47.2	47.1	47.1	47.1		47.1	47.1	
of which : grants	4.5	9.2	6.1			8.3	8.3	8.3	8.3	8.3	8.3		8.3	8.3	
Primary (noninterest) expenditure	42.4	43.9	44.5			44.2	44.0	43.8	43.8	43.9	43.9		44.0	44.0	
Automatic debt dynamics	9.8	3.5	-1.7			-1.8	-1.7	-1.1	-1.1	-1.0	-0.9		-0.2	0.1	
Contribution from interest rate/growth differential	9.1	3.2	-2.3			-2.4	-2.2	-1.5	-1.4	-1.3	-1.2		-0.4	0.0	
of which : contribution from average real interest rate	4.5	3.2	2.1			1.0	1.0	1.5	1.5	1.4	1.4		1.7	1.2	
of which : contribution from real GDP growth	4.6	0.0	-4.4			-3.4	-3.2	-3.0	-2.9	-2.8	-2.7		-2.1	-1.2	
Contribution from real exchange rate depreciation	0.7	0.3	0.6			0.6	0.4	0.4	0.4	0.3	0.3		...	...	
Other identified debt-creating flows	0.0	0.0	-5.6			-2.5	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			-2.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	-0.9			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other) 6/	0.0	0.0	-4.6			-0.6	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	22.6	5.9	-2.4			0.0	-0.2	0.1	0.1	0.1	0.1		0.1	0.1	
NPV of public sector debt	...	...	90.0			83.3	79.9	77.7	75.4	73.2	70.9		59.3	34.0	
o/w foreign-currency denominated	...	...	54.6			55.3	53.6	51.1	48.7	46.2	42.5		29.7	17.4	
o/w external	...	...	54.6			55.3	53.6	51.1	48.7	46.2	42.5		29.7	17.4	
NPV of external debt in percent of exports	...	...	115.0			111.8	108.2	102.9	97.9	92.7	85.0		59.4	34.8	
External debt service in percent of exports	11.8	19.5	20.8			6.1	8.4	10.0	9.6	9.5	11.9		7.2	3.8	
NPV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	7.0	4.6	2.6			1.8	2.6	3.1	3.0	4.0	5.8		3.7	1.8	
NPV of public sector debt-to-revenue ratio (in percent) 3/			184.4			175.5	169.1	164.8	160.2	155.4	150.5		126.0	72.2	
o/w external			111.7			116.5	113.5	108.4	103.5	98.2	90.2		63.1	37.0	
Debt service-to-revenue ratio (in percent) 3/ 4/	14.2	21.3	14.3			10.6	12.5	13.7	13.3	15.3	19.1		14.4	10.4	
Primary deficit that stabilizes the debt-to-GDP ratio	-32.4	-9.4	9.6			4.3	1.9	1.0	1.0	0.9	0.8		0.2	-0.2	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	-4.7	0.0	3.5	0.9	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Average nominal interest rate on forex debt (in percent)	5.0	3.9	4.0	3.7	1.0	2.1	2.1	2.8	2.7	2.7	2.7	2.5	2.8	3.2	2.9
Average real interest rate on domestic currency debt (in percent)	8.0	3.3	1.4	6.1	3.0	3.5	3.7	3.8	3.8	3.7	3.8	3.7	4.6	5.4	4.7
Real exchange rate depreciation (in percent, + indicates depreciation)	1.0	0.4	0.7	0.3	1.0	0.8	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	0.6	1.5	1.4	1.5	0.8	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Growth of real primary spending (deflated by GDP deflator, in percent)	6.0	4.5	5.0	2.7	11.4	2.1	2.5	2.5	2.8	2.9	2.9	2.6	2.9	2.8	2.9
Grant element of new external borrowing (in percent)	...	...	...			35.6	35.6	35.6	35.6	35.6	35.6	35.6	35.6	35.6	...

Sources: Country authorities; and Fund staff estimates and projections.

1/ Non-financial Public Sector (includes debt with Dominica's Social Security System)

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues including grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

6/ In 2004, it is assumed that all nonparticipating creditors received the intermediate bond, which carries a face value reduction of 20 percent.

7/ Refers to external debt. Assumes that nonpartcipating creditors receive the intermediate bond. Excludes external interest arrears. Arrears with participating creditors have been settled as part of the debt restructuring. Arrears with nonparticipating creditors are either in dispute or expected to be settled when an agreement is reached. Undisputed interest arrears are approximately 0.4 percent of GDP

8/ Fiscal year aggregates presented on a fiscal year basis

Table 2a.Dominica: Sensitivity Analysis for Key Indicators of Public Debt 2005-2025

	Estimate	Projections						
	2005	2006	2007	2008	2009	2010	2015	2025
NPV of Debt-to-GDP Ratio								
Baseline	83	80	78	75	73	71	59	34
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	83	86	90	94	99	104	133	229
A2. Primary balance is unchanged from 2004	82	78	74	71	67	63	43	-8
A3. Permanently lower GDP growth 1/	83	81	79	78	77	76	75	94
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2006-2007	83	86	92	94	96	98	109	140
B2. Primary balance is at historical average minus one standard deviations in 2006-2007	83	89	95	94	92	90	80	54
B3. Combination of B1-B2 using one half standard deviation shocks	83	89	97	95	92	90	75	38
B4. One-time 30 percent real depreciation in 2006	83	104	102	100	98	95	81	45
B5. 10 percent of GDP increase in other debt-creating flows in 2006	83	90	88	86	84	83	74	55
NPV of Debt-to-Revenue Ratio 2/								
Baseline	175	169	165	160	155	150	126	72
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	175	181	190	198	207	216	272	449
A2. Primary balance is unchanged from 2004	173	164	158	150	143	135	91	-17
A3. Permanently lower GDP growth 1/	175	170	168	165	163	161	158	197
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2006-2007	175	181	192	196	200	204	227	295
B2. Primary balance is at historical average minus one standard deviations in 2006-2007	175	187	202	199	195	191	170	115
B3. Combination of B1-B2 using one half standard deviation shocks	175	188	204	199	194	188	159	81
B4. One-time 30 percent real depreciation in 2006	175	220	216	212	207	202	173	96
B5. 10 percent of GDP increase in other debt-creating flows in 2006	176	190	187	183	180	176	157	116
Debt Service-to-Revenue Ratio 2/								
Baseline	11	12	14	13	15	19	14	10
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	11	13	17	19	23	29	33	61
A2. Primary balance is unchanged from 2004	11	12	12	12	13	17	10	-2
A3. Permanently lower GDP growth 1/	11	13	14	14	16	20	19	27
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2006-2007	11	13	16	18	22	27	27	40
B2. Primary balance is at historical average minus one standard deviations in 2006-2007	11	12	20	23	21	24	19	16
B3. Combination of B1-B2 using one half standard deviation shocks	11	13	20	22	20	23	17	11
B4. One-time 30 percent real depreciation in 2006	11	13	16	16	18	21	16	10
B5. 10 percent of GDP increase in other debt-creating flows in 2006	11	12	21	17	18	22	18	16

Sources: Country authorities; and Fund staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
2/ Revenues are defined inclusive of grants.